CORE PURE / PROOF BY INDUCTION

Proof by Induction - Divisibility

Mark scheme 8 questions · 48 marks

Questions

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	Total	**48**	

Reading the mark scheme

These rules apply throughout, unless a note says otherwise.

Marks

Every mark is worth one, so **M1 A1** on a row is two marks. Only marks coded ft follow through.

Code	Meaning
M1	Method: a correct method, applied to this question. A slip doesn't cost it if the method is clear. A formula must be used, not just quoted.
dM1	Dependent method: as M1, but only if the method mark before it was earned, unless its note names another.
A1	Accuracy: a correct answer or step. It needs the method mark it follows.
A1ft	As A1, following through from an earlier answer that the note names.
A1 AG	The last mark of a part whose answer is printed in the question: complete working, with no errors.
B1	A correct value, fact or statement. It needs no method mark.
B1ft	As B1, following through from an earlier answer that the note names.
E1	A written explanation, interpretation or conclusion. Concluding a proof by induction needs every earlier mark.

Abbreviations

ft and AG qualify a code in the Marks column. The others appear in the notes.

ft	follow through	**AG**	answer given in the question
cao	correct answer only, with no follow-through	**cso**	correct solution only: no errors in the part
oe	or an equivalent form	**awrt**	any answer that rounds to this value
soi	seen, or implied by later working	**isw**	ignore working after a correct answer
SC	special case: credit outside the main method	**their**	the student's own earlier value
condone	a slip that doesn't cost the mark	[ ]	working the mark doesn't need
s.f.	significant figures	**d.p.**	decimal places

Reading the scheme

- Each row is a line of working, with the marks it earns beside it. Where a row carries two marks, each note starts with its code.
- [$\boldsymbol{n}$] ends each part with its total.
- A shaded *Alternative* is a second method for the same total. Mark an answer by the method it follows.
- Any other valid method earns the equivalent marks, unless the question names a method or says *hence*.
- A sketch earns one B1 for each feature named below its figure.

Accuracy and answers

- Unless a question says otherwise, give answers to 3 significant figures, and angles in degrees to 1 decimal place. Angles in radians take 3 significant figures.
- A final answer with more figures is accepted if it rounds to the scheme's value; one with fewer is not. An answer spoiled by rounding too early loses its accuracy mark.
- Where a question asks for an exact answer, a decimal loses the final accuracy mark. Otherwise either is accepted.
- Wrong or missing units cost nothing unless the question asks for them.
- An omitted $\mathrm{d}\theta$, $\mathrm{d}x$ or $\mathrm{d}u$ in an integral is not penalised where the variable of integration is clear from the integrand and the limits. It is penalised where it is genuinely ambiguous: a substitution part-way through, two variables in play, or limits in the wrong variable.
- A correct answer with no working earns full marks, unless the question needs the working: show that, prove, hence, a method it names, or an exact answer. A wrong answer with no working earns nothing. Calculator answers are treated the same way.
- A correct answer, once reached, isn't spoiled by later working (isw).

Value of $\boldsymbol{g}$

- Take $g = 9.8\,\mathrm{m\,s^{-2}}$ unless a question states a value, which is then the only one accepted. Answers that use $g = 9.8$ may be given to 2 or 3 significant figures.
- Where a question doesn't state g, an answer worked with $g = 10$ throughout earns the same marks, keeping any value printed in the question. The worked solutions list the final answers with $g = 10$.

Question	Scheme	Marks	Notes
1	$n = 1:\ 1^5 - 1 = 0$, divisible by 5	**B1**	Needs the value 0 and a statement that it is divisible by 5, or true for $n = 1$.
	Assume true for $n = k$: $k^5 - k = 5m$ Considers $(k+1)^5 - (k+1)$	**M1**	States the assumption that $k^5 - k$ is divisible by 5 and considers the expression with $n = k + 1$.
	$(k+1)^5 - (k+1) - (k^5 - k)$ $= (k^5 + 5k^4 + \ldots + 1) - (k+1) - (k^5 - k)$	**M1**	Expands $(k+1)^5$ to six terms, condoning coefficient slips, and forms the difference, or another multiple of $k^5 - k$ subtracted, or rearranges to isolate $k^5 - k$.
	$= 5k^4 + 10k^3 + 10k^2 + 5k$	**A1**	Or the correct equivalent expression for their route.
	$(k+1)^5 - (k+1)$ $= (k^5 - k) + 5(k^4 + 2k^3 + 2k^2 + k)$ $\Rightarrow$ divisible by 5	**A1**	cso. Needs the factor 5 shown and the assumption used for $k^5 - k$.
	True for $n = 1$; true for $n = k$ $\Rightarrow$ true for $n = k + 1$ $\Rightarrow$ true for all positive integers n	**E1**	States true for $n = 1$, true for $n = k$ implies true for $n = k + 1$, hence true for all positive integers n.
	Total for Question 1	**6**	

Question	Scheme	Marks	Notes
2	$u_1 = 5^2 - 2^1 = 23$	**B1**	u_1 evaluated as 23, or $25 - 2$; "true for $n = 1$" alone is B0.
	Assume $5^{2k} - 2^k = 23m$ $u_{k+1} = 5^{2(k+1)} - 2^{k+1}$	**M1**	Assumes u_k divisible by 23 and writes u_{k+1}, or $u_{k+1} - \lambda u_k$, in terms of k.
	$= 25 \times 5^{2k} - 2 \times 2^k$	**M1**	Uses index laws to write u_{k+1} in terms of 5^{2k} and 2^k. Condone one coefficient slip.
	$= 23 \times 5^{2k} + 2\left(5^{2k} - 2^k\right)$	**A1**	oe, e.g. $25\left(5^{2k} - 2^k\right) + 23 \times 2^k$, or $u_{k+1} - 2u_k = 23 \times 5^{2k}$.
	$= 23 \times 5^{2k} + 2(23m)$ $= 23\left(5^{2k} + 2m\right)$	**A1**	cso. Uses the hypothesis to show u_{k+1} is a multiple of 23: factor 23 explicit, or each term stated divisible by 23.
	True for $n = 1$; true for $n = k \Rightarrow$ true for $n = k + 1$; so true for all $n \in \mathbb{Z}^+$	**E1**	States all three points: true for $n = 1$, truth for $n = k$ implies truth for $n = k + 1$, so true for all positive integers n.
	Total for Question 2	**6**	

Question	Scheme	Marks	Notes
3	$n = 1: \ 9 - 8 - 1 = 0$, divisible by 64	**B1**	Needs the value 0 and a statement that it is divisible by 64, or $0 = 64 \times 0$.
	Assume $9^k - 8k - 1 = 64m$ Considers $9^{k+1} - 8(k+1) - 1$	**M1**	States the assumption for $n = k$ and writes the expression for $n = k + 1$. Not for assuming the result for $n = k + 1$.
	$9^{k+1} - 8(k+1) - 1 = 9 \times 9^k - 8k - 9$ $= 9(9^k - 8k - 1) + \ldots$	**M1**	Uses $9^{k+1} = 9 \times 9^k$ and relates the result to $9^k - 8k - 1$ or $64m$, e.g. by substituting $9^k = 64m + 8k + 1$ or forming a difference with the $n = k$ expression. Condone slips.
	$= 9(9^k - 8k - 1) + 64k$	**A1**	Correct expression, oe such as $9(64m + 8k + 1) - 8k - 9$.
	$= 9(64m) + 64k$ $= 64(9m + k)$	**A1**	cso. Shown as 64 times an integer expression, or both terms identified as multiples of 64.
	True for $n = 1$; true for $n = k \Rightarrow$ true for $n = k + 1$ so true for all positive integers n	**E1**	Concludes, naming: true for $n = 1$; true for $n = k$ implies true for $n = k + 1$; so true for all positive integers n.
	Total for Question 3	**6**	

Question	Scheme	Marks	Notes
4	$n = 0$: $2^0 + 2 \times 3^0 + 4^1 = 7$	**B1**	Evaluation to 7 shown. Checking $n = 1$ alone does not score.
	Assumes $2^k + 2 \times 3^{2k} + 4^{2k+1} = 7m$ $n = k + 1$: $2^{k+1} + 2 \times 3^{2k+2} + 4^{2k+3}$	**M1**	States the assumption for $n = k$ and writes the expression with $n = k + 1$ substituted.
	$= 2 \times 2^k + 18 \times 3^{2k} + 16 \times 4^{2k+1}$	**M1**	Writes each term of the $n = k + 1$ expression, or of a combination such as its difference from the $n = k$ expression, as a multiple of 2^k, 3^{2k} or 4^{2k+1}. Condone a coefficient slip.
	$= 2\left(2^k + 2 \times 3^{2k} + 4^{2k+1}\right)$ $+ 14 \times 3^{2k} + 14 \times 4^{2k+1}$	**A1**	oe: a multiple of the $n = k$ expression plus terms with a visible factor 7, or with one power replaced using the assumption.
	$= 14m + 14\left(3^{2k} + 4^{2k+1}\right)$ $= 7\left(2m + 2 \times 3^{2k} + 2 \times 4^{2k+1}\right)$	**A1**	cso. The $n = k + 1$ expression as 7 times an integer expression, or a sum of multiples of 7, with the assumption applied.
	True for $n = 0$ True for $n = k \Rightarrow$ true for $n = k + 1$ $\Rightarrow$ true for all non-negative integers n	**E1**	States true for $n = 0$, true for $n = k$ implies true for $n = k + 1$, so true for all non-negative integers n. Condone "for all n".
	Total for Question 4	**6**	

Question	Scheme	Marks	Notes
5	$n = 1 : 7^1 - 6(1) - 1 = 0$, divisible by 36	**B1**	Needs the value 0 and a statement that it is divisible by 36, or $0 = 36 \times 0$.
	Assume $7^k - 6k - 1 = 36m$ $n = k + 1 : 7^{k+1} - 6(k+1) - 1$	**M1**	States the assumption for $n = k$, in words or as $= 36m$, and writes the expression with $n = k + 1$.
	$7^{k+1} - 6(k+1) - 1 = 7 \times 7^k - 6k - 7$ $= 7(7^k - 6k - 1) + \ldots$	**M1**	Writes 7^{k+1} as 7×7^k and expresses the $n = k + 1$ expression using $7^k - 6k - 1$ or $36m$, or subtracts a multiple of it. Condone slips in the remainder.
	$= 7(7^k - 6k - 1) + 36k$	**A1**	Correct unsimplified expression, oe such as a correct difference equal to $36k$.
	$= 7(36m) + 36k = 36(7m + k)$	**A1**	Factor of 36 shown, or every term stated to be a multiple of 36, using the assumption.
	True for $n = 1$; true for $n = k \Rightarrow$ true for $n = k + 1$; so true for all $n \in \mathbb{Z}^+$	**E1**	States true for $n = 1$, true for $n = k$ implies true for $n = k + 1$, and hence true for all positive integers n.
	Total for Question 5	**6**	

Question	Scheme	Marks	Notes
6	$f(1) = 9 - 8 + 63 = 64$	**B1**	64 must be seen; $f(1)$ stated divisible by 64 without its value is B0.
	Assume $9^k - 8k + 63 = 64m$ $f(k+1) = 9^{k+1} - 8(k+1) + 63$	**M1**	States the assumption for $n = k$ and substitutes $k + 1$ into f. Not for assuming the result for $n = k + 1$.
	$f(k+1) = 9 \times 9^k - 8k + 55$ $= 9(9^k - 8k + 63) + \ldots$	**M1**	Writes 9^{k+1} as 9×9^k and expresses $f(k + 1)$ in terms of $f(k)$, or forms $f(k + 1) - 9f(k)$ oe. Condone arithmetic slips.
	$f(k+1) = 9(9^k - 8k + 63) + 64k - 512$	**A1**	oe, unsimplified accepted, such as $f(k + 1) - 9f(k) = 64k - 512$.
	$f(k+1) = 9(64m) + 64(k - 8)$ $= 64(9m + k - 8)$	**A1**	cso. Every term shown to be a multiple of 64, with $f(k + 1)$ stated divisible by 64.
	True for $n = 1$; true for $n = k \Rightarrow$ true for $n = k + 1$; so true for all $n \geq 1$	**E1**	States all three points: true for $n = 1$, true for k implies true for $k + 1$, so true for all $n \geq 1$.
	Total for Question 6	**6**	

Question	Scheme	Marks	Notes
7	$n = 1$: $7^2 + 8 \times 13 - 9 = 144 = 36 \times 4$	**B1**	Shows 144 and that it is a multiple of 36.
	Assumes $7^{2k} + 8 \times 13^k - 9 = 36m$ $n = k + 1$: $7^{2k+2} + 8 \times 13^{k+1} - 9$	**M1**	States the assumption for $n = k$ and writes the expression with $n = k + 1$. Condone divisibility assumed in words.
	$= 49 \times 7^{2k} + 104 \times 13^k - 9$ $= 13(7^{2k} + 8 \times 13^k - 9) + \ldots$	**M1**	Uses index laws and rearranges in terms of the $n = k$ expression, or forms the $n = k + 1$ expression minus a multiple of it. Condone arithmetic slips.
	$= 13(7^{2k} + 8 \times 13^k - 9) + 36(7^{2k} + 3)$	**A1**	Correct, unsimplified, oe such as $49(7^{2k} + 8 \times 13^k - 9) - 36(8 \times 13^k - 12)$.
	$= 36(13m + 7^{2k} + 3)$	**A1**	Uses the assumption to show a multiple of 36, by a factor of 36 shown or each term stated divisible by 36.
	True for $n = 1$; true for $n = k$ $\Rightarrow$ true for $n = k + 1$; so true for all positive integers n	**E1**	Concludes with all three: true for $n = 1$, true for $n = k$ implies true for $n = k + 1$, and so true for all positive integers n.
	Total for Question 7	**6**	

Question	Scheme	Marks	Notes
8	$n = 1 : 13 + 12 + 11 = 36 = 12 \times 3$	**B1**	Needs 36 shown as a multiple of 12, or stated divisible by 12.
	Assumes $13^k + 3 \times 4^k + 11 = 12m$ Considers $13^{k+1} + 3 \times 4^{k+1} + 11$	**M1**	States the assumption for $n = k$ and writes the expression for $n = k + 1$. Not for the $n = k + 1$ expression alone.
	$(13^{k+1} + 3 \times 4^{k+1} + 11)$ $\quad - (13^k + 3 \times 4^k + 11)$ $= 13^k(13 - 1) + 3 \times 4^k(4 - 1)$	**M1**	Subtracts the $n = k$ expression, or any multiple of it, from the $n = k + 1$ expression and writes 13^{k+1}, 4^{k+1} in terms of 13^k, 4^k. Condone slips.
	$= 12 \times 13^k + 9 \times 4^k$	**A1**	Or the correct simplified result for another multiple subtracted.
	$13^{k+1} + 3 \times 4^{k+1} + 11$ $\quad = 12m + 12 \times 13^k + 12(3 \times 4^{k-1})$ $\quad = 12(m + 13^k + 3 \times 4^{k-1})$	**A1**	cso. Shows 9×4^k as a multiple of 12 and uses the hypothesis to write the $n = k + 1$ expression as 12 times an integer.
	True for $n = 1$; true for $n = k \Rightarrow$ true for $n = k + 1$ so true for all $n \in \mathbb{Z}^+$	**E1**	Concludes with true for $n = 1$, true for $n = k$ implies true for $n = k + 1$, and so true for all positive integers n.
	Total for Question 8	**6**	